

IT460 Fish Game

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Introduction - Concept

Shipwreck on a trash island, there was another with you but as you approached the unconscious body. A fish a medium size of a dog, pounce out of the water fins leg like as it bites and drag the man into the water. Starting from the pain, the man startles and screams in agony as he is dragged down below. It would be a good idea to not be so close to the water, or at least a beach where fish can come up and bite you. You traverse a little further inland, finding traces of others that might exist here on the island. You end up on a cliff edge, with a hut, a skeleton, and a journal. A grim warning, the only way to survive is to fish, but fish are mutated and they will fight back.

Intro sequence

- Shipwreck
- Washed ashore
- Characters witness an npc dragged into the water by a fish with jaws and fins forearms.
- Explore up the coast
- Finds hut cliff side
- Journal as a guide for new notes and unlock
- Starts fishing and fighting fish to survive
- If a player dies, tribal mutated platypuses will bring them back to the hut.

- Eventually the idea to fish up trash god, slay it to actually be able to leave the island.

Gameplay - Play Loop

I really like how Monster hunter wilds did their fishing minigame. Walking around in a third person view, but then the camera can focus in for fishing. Fishing will be the main focus on the zoom in mode, where the player model becomes the background. With the purpose of making the game as realistic as possible, the player will just have a rod and a hook, not able to see the fish in the water at all. Fish, fight, loot, craft and repeat is the game loop I am going for. Since fish are mutated, and the theme is trash, fishing and slaying fish will give resources to upgrade gear, and some fish can also drop gear.

Gameplay - Setting

Trash island, surrounded by spires and spikes that look like claws/fangs reach up into the sky. The island itself should be a mixture of sand and garbage, but mainly garbage. Dark stormy clouds with drizzles sometime, purple sludge and crystals everywhere for dim lighting.

Backend - Engine - Code

For the Trash Game project, Godot and its coding language is selected. With experience working with 2d game dev with Godot, a 3d game would be the natural next thing I want to try. Godot has a 3d engine that flows really well if you have experience with its 2d game dev. Compared to other engines, Godot is meant for 2d games with a lighter asset library for 3d. Unreal and Unity are the other options, Unreal offering way too much of what I need and is looking for. Unity is beginner friendly and it has support for more targeted platforms. Godot is open source, and the price of free is hard to beat, plus I already have some experience with Godot and how it wants to organize itself.

Unreal Games

- Fortnite: The global battle royale phenomenon.
- Black Myth: Wukong: An action RPG renowned for its top-tier visual quality.
- Star Wars Jedi series: High-performance modern action-adventure games.
- Final Fantasy VII Remake: A massive JRPG demonstrating high-fidelity character rendering.

Unity Games

- Among Us: A massively popular 2D multiplayer social deduction game.
- Pokémon GO: A leading example of large-scale augmented reality (AR) gaming.
- Hollow Knight: A critically acclaimed 2D indie Metroidvania.

- Cuphead: Known for its unique, hand-drawn 1930s cartoon art style.
- Genshin Impact: A major cross-platform open-world RPG (developed in Unity, though often mistaken for proprietary tech).

Godot Games

- Buckshot Roulette: A viral indie horror game known for its atmospheric 3D design.
- Brotato: A highly successful top-down arena shooter roguelite.
- Cassette Beasts: An open-world RPG that showcases Godot's capability for complex 2D systems.
- Dome Keeper: A unique roguelike mining game that became a significant hit on Steam.
- Slay the Spire 2: The upcoming sequel to the legendary deck-builder was moved to Godot for development.

Backend - File Organization

1. Res

a. Asset

i. Models

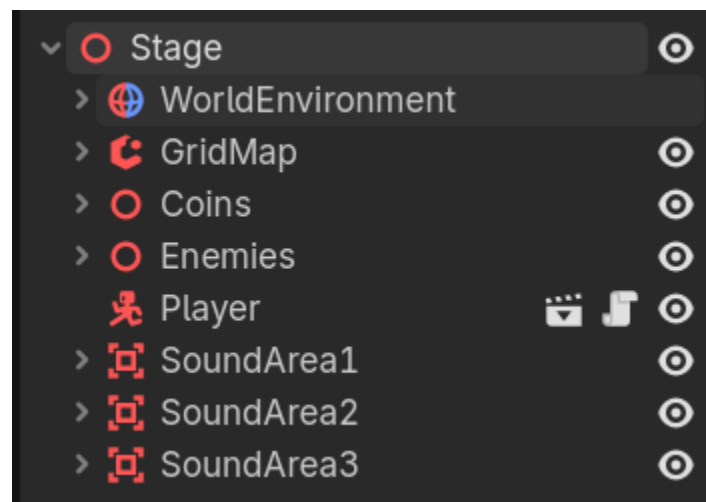
1. Mob
2. Players
3. Items
4. Equipment

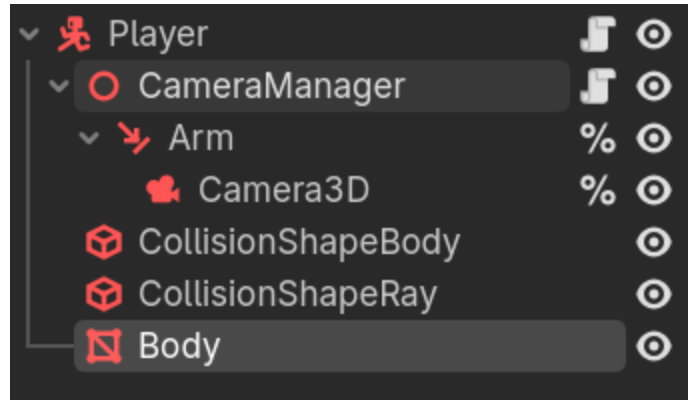
ii. World

1. Land
2. Structure

b. Root

- Stage = World
- WorldEnvironment = Lighting vibe
- GridMap = Physical map and tiles that sets the stage.





Backend - Models - Blender

There is concern here for using blender, it has been stated that it has a tougher learning curve for beginners. With that said I have also messed around with Blender before, not to the extent I would like but better than nothing. Blender is also a free and open source option to use for 3d modeling.

The models and assets that I will be doing won't be high definition, which is one of the reasons why Unreal engine wasn't chosen. Graphics is not the purpose of this game, the graphics and aesthetic that I'm thinking of and looking for is closer to Peak RV there yet, a grim dark fishing horror game with style and aesthetics as Pokopia.

Backend - Models - Terrain

Backend - Models - Terrain - Terrain3D

Third party plug in that lets Godot model terrain needed for creation of trash island. Written in C++

Backend - Models - Terrain - Water

Godot shaders provide options for water in the game. PSX Style Water Surface – Pixelation, Waves, Scrolling Textures was selected for the water in the game, and the island was redone a little more, to compensate for scaling and water.

Notes

Navigation3d is an experimental node, it offers the ability to overlap regions.

Objects in game requires

- Node3d - The object itself
 - Meshinstance3d
 - CollisionShape3d

Fishing Pole - a 3d node of it's own

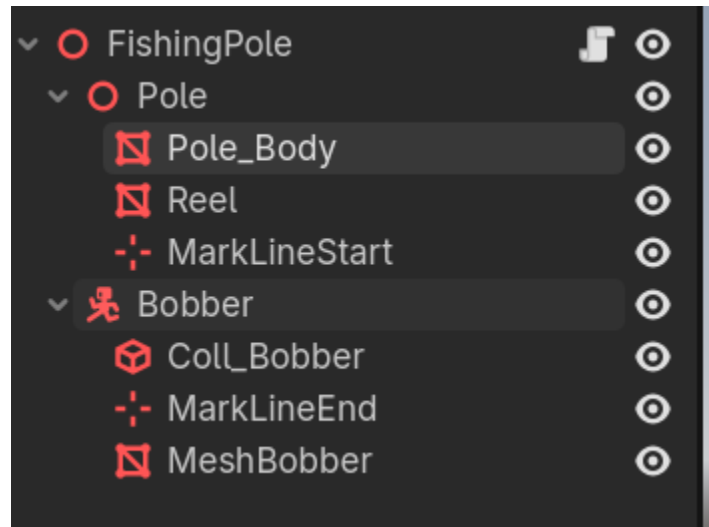
- Pole - the pole itself, and that's what the animation player is attached to. As

Cast_Pole.

- Has two mesh one for the pole body itself and the other is the reel.
 - Since multiple mesh can be attached to one 3d node that means the softbody3d is just a meshinstance3d for the Pole's body.
- Bobber (CharacterBodyNode)- Bobber is its own separate child, so it doesn't have to be attached to the pole. Bobber is a Character Body 3d, it needs its own script on how it behaves.
 - Line doesn't show up
 - Due to how the line and bobber interacts
 - Need to do a reeling animation

Softbody3d consumes way too much resources for it to be used as a simple fishing line.

CharacterBody can be moved by physics and it's kinematic, it can collide with physics objects but will not be moved and automates mechanics like ramps.



Files for the FishingPole

SpringArm3D casts a ray or a shape along its Z axis and moves all its direct children to the collision point, with an optional margin.

Height map - godot for elevation and water.

Spawn manager

- Rates
- Variation
 - Size - Change of scale
 - Color - Atlas shaders for the fish

Gameplay

- Player
 - When cast, enters state fishing
 - Locks wasd movement
 - Cam fix on bobber
- Bobber
 - Bobber lands and there be an indicator if there are fish near by, a hot or cold mechanic
 - Bobber's circular area, when it lands and starts bobbing in the Ocean, are there fish hotspot markers nearby. There could be multiple markers inside the area of the bobber, it needs to find the closest one.
 - Add Area3D to markers so there is something for the bobberareacol to hit
 - Hotspots are randomized per 30 seconds, the randomized method is not being called and it should be called in the hotspot manager with the usage of a timer node. Keep inside the hotspot manager instead, call the timer node for the timing.
 - We know that the hotspots are working and they are being randomized.

- Update `_fishing_state` for player this is where the actual gameplay begins.
 - So what happens if there is a bite?
 - The bobber can disappear
 - The camera still fix to the bobber but the bobber no longer has a visual like turning off the mesh so it's unseen.
 - Then the camera should follow the invis bobber as if the fish is swimming left to right.
 - Player needs to press a or d depending on if the fish is going left of right. Left is A then D is right.
 - After some time the fish should be tired and that's when the player can reel by pressing wasd in a circular pattern like spinning the handle on a reel. Then after some time the fish should start it's left and right behavior again.
 - Casting bobber into the water, there will be few rings of possible fish spots, closer to the center of those invisible rings, the stronger the catch rate.
 - 5 invisible circles will spawn and stay in a specified area, for about 15 seconds before randomly relocating in the area.
 - On bit, an indicator will appear to let the player know to set the hook.

- The bobber will be hidden, then splashing on the water telling the player which fish is swimming.
 - The goal is to keep the tip of the pole ahead of the fish's swimming direction.
 - Once the fish is tired and stops swimming, the player can reel the fish in by alternating clicks left and right click.
- Sprite3D = It will be cast, it will only appear on top of the Bobber and only while the bobber is inside the FishingZone. if it's close(white), closer(yellow), or closest(Orange) to a fish. Then every time there's a bite check the icon will shake, it should look like a pin on a map.
 - Once there's a bite, it should transform to a live button guide ui that is above the fish. With two keys on it a and d, it blinks which server the player should press to tire the fish. It will shake a blink red when the wrong key is pressed.
 - Once the fish is tired it will turn into four keys wasd always in the a circular motion of wdsa for reeling, then blink in the pattern the player is supposed to press to reel in the fish, if the wrong button is press, it resets and starts at w and blinks red.

FishingZone (Area3D) - Is it's own scene and it has three collision shapes all fixed at y 15 in position around the island where the player can fish.

- FishingZoneMain
 - Shuffles and keeps 7 HotSport markers, at y 15, and within the zone.
 - Functions
 - `_mix_hotspot` that is called on ready, and then every 30 seconds unless fish are currently biting. Then that timer is paused.
 - `_get_random_point_in_zone`
 - The markers are meant for the bobber to calculate how close it is, for an indicator of hot or cold. Telling the player how close they are to a fish, while also calculating bite chance.

Player.gd

- `_ready`
 - Sets up signal connections, and initialization
- `_unhandled_input`
 - Handles pov range of camera.
- `_physics_process`
 - Constant processing of physics, such as player state in game.

- `_roaming_state`
 - While not fishing, players can move around and jump.
- `_fishing_state`
 - Locks wasd movement, camera should start watching bobber.
- `_enter_fishing_state`
 - Transition into fishing
- `_exit_fishing_state`
 - Exiting fishing
- `_cast`
 - Play cast animation and start of fishing
- `_apply_gravity`
 - Exist in case other states were to be introduced such as diving.
- `_handle_jump`
 - Isolated jump physics in case swimming and diving is introduced to lock players out of movements.
- `_handle_movement`
 - Isolated to lock character movements per state.
- `_toggle_free_cam`
 - Set up for UI interactions

Game UI

- When root initialize, we'll have the island in the back ground, with it's waves crashing.

- Menu options game home page
 - Loads in game - players have 3 minutes to catch as many fish as they can and there will be a score on the upper right.
 - There are three tiers of difficulty, points goes as low = 100, medium = 250, then hard 400 points. The lesser the time it takes to reel in there is a percentage 5-15% score boost to that catch.
 - How to play
 - Windows pop up
 - “Cast, see if there are fish nearby, wait for a bite. Press A or D depending on which way the fish goes. Then once the fish is tired, reel in through spamming wdsa till the fish is caught. You got 3 minutes go nuts.”
 - Quit - closes the app
- InGameUI - Has a timer at the top center of screen 3 min counting down, then on the upper right the total score.
 - When the 3 minutes is up a game window will pop up, then total score and a button to return to the main menu. Where the player can click play to start a new game of fishing.

Resources Used

SRS Doc

<https://www.perforce.com/blog/alm/how-write-software-requirements-specification-srs-document#what-is-a-software-requirements-specification-document-01>

Godot 3d Game demo

<https://github.com/gdquest-demos/godot-4-3d-third-person-controller>

Tempt player model

<https://quaternius.itch.io/universal-animation-library>

Godot set up instructions

<https://quaternius.itch.io/universal-animation-library/download/eyJpZCI6MzQwODAzNCwiZXhwaXJlcyI6MTc3NjcwMzM2MH0%3d%2eS5YY%2bywb0KqRiz4fyh74BO4f1MA%3d>

Tempt fish model

<https://alstrainfinite.itch.io/fish>

Water Shaders

<https://godotshaders.com/shader/psx-style-water-surface-pixelation-waves-scrolling-textures/>

Terrain3D Documentation

<https://terrain3d.readthedocs.io/en/stable/docs/introduction.html>

Colton Sky Shaders

<https://cdn.discordapp.com/attachments/1252366705930276928/1509243165624897546/GSHSky.gdshader?ex=6a1f0f3b&is=6a1dbdbb&hm=0d52fb5d137a9dd197eb32ac010c5d5879cfc2fb92378b48a16ecea0e1355e33&>

Texture Assets

<https://binbun3d.itch.io/godot-grass/download/eyJleHBpcmVzIjoxNzgwMjcyMzk2LCJpZCI6NDI2OTc0OH0%3d%2e1W8nTh2GLafH6ZZvDQuh3UaNGxg%3d>

Sprite Keys

<https://tekilor.itch.io/2d-ui-keyboard>

Godot Forum

<https://godotforums.org/>

Fishing Line Set Up Tutorial

<https://www.youtube.com/watch?v=U9K2wvZslWc>